

I: THE SIX PILLARS OF DATA INTEGRITY

1.1 The "Garbage In, Garbage Out" (GIGO) Principle

If the input data is flawed, every subsequent Machine Learning model, Big Data analytic, or business report becomes unreliable. Poor data quality can lead to biased AI, incorrect financial forecasting, and operational failures.

1.2 The 6 Core Dimensions of Data Quality

To objectively measure the health of your data, we use these global standards:

- **Accuracy:** Does the data correctly reflect the real-world event? (e.g., A delivery date cannot be before the order date).
- **Completeness:** Are there critical missing values (Nulls)? We check if mandatory fields like "Customer ID" or "Transaction Amount" are populated.
- **Consistency:** Is the data uniform across different systems? (e.g., A user's status shouldn't be "Active" in the CRM but "Inactive" in the Billing system).
- **Timeliness:** Is the data up-to-date? For real-time Big Data, data that is even a few hours old might be obsolete.
- **Validity:** Does the data follow the required format or schema? (e.g., Phone numbers must only contain digits; Email addresses must include an '@' symbol).
- **Uniqueness:** Are there duplicate records? Redundant data skews statistical averages and wastes storage space.

II: TECHNICAL VALIDATION & CLEANING WORKFLOW

2.1 Schema and Type Validation

Before the data reaches the processing engine (Spark/Kafka), it must pass the "Shape Test":

- **Data Type Enforcement:** Ensuring numeric columns are not stored as strings.
- **Constraint Checks:** Verifying that values fall within logical ranges (e.g., Age: 0–120; Probability: 0–1).
- **Cross-Field Validation:** Checking logic between columns (e.g., If "Status" is "Married", the "Spouse Name" field should not be empty).

2.2 Automated Data Profiling

Using tools like **Pandas Profiling** or **Great Expectations**, the system automatically generates a "Data Health Report" including:

- **Record Count:** Total rows vs. expected rows.
- **Distribution Analysis:** Mean, Median, and Standard Deviation to identify anomalies.
- **Outlier Detection:** Utilizing Z-Score or IQR (Interquartile Range) to flag extreme values that might be errors.

2.3 Handling Missing Data (Imputation vs. Deletion)

We define the strategy for missing values:

- **Dropping:** Removing rows with missing critical data.
- **Imputation:** Filling gaps with the mean, median, or a predicted value based on other features.

III: MONITORING & GOVERNANCE FRAMEWORK

3.1 Data Quality Service Level Agreements (SLAs)

We establish "Acceptance Thresholds" for the pipeline:

- **Green (Healthy):** Error rate $< 0.5\%$ — Proceed to training/analysis.
- **Yellow (Warning):** Error rate $0.5\% - 2\%$ — Proceed but flag for manual review.
- **Red (Critical):** Error rate $> 2\%$ — Stop the pipeline; trigger data re-extraction.

3.2 Real-time Monitoring and Alerting

In a production cloud environment, the checklist is automated:

- **Dashboards:** Visualizing data health scores over time.
- **Slack/Email Alerts:** Immediate notification to the Data Engineering team if a source feed becomes "noisy" or "dirty."

3.3 Data Lineage and Audit Trails

If an error is found, we must be able to trace it back to the source.

- **Traceability:** Mapping how data moved from the source (S3/SQL) through the cleaning steps to the final model.
- **Versioning:** Keeping snapshots of the data quality at different points in time.